# Geometry: Trapezoid Midsegment Theorem

Practice Worksheet • numberbender.com

Name: ______________________ Date: ______________ Score: / 8

**DIRECTIONS** *MN is the midsegment of trapezoid ABCD, connecting the midpoints of the legs. Apply the Midsegment Theorem: MN = (b1 + b2) / 2. Show all work.*

**1** Find MN:

$b_1 = 7, \quad b_2 = 15$

Answer: ______________________

**2** Find MN:

$b_1 = 5, \quad b_2 = 23$

Answer: ______________________

**3** Find b2:

$MN = 13, \quad b_1 = 9$

Answer: ______________________

**4** Find b1:

$MN = 16, \quad b_2 = 10$

Answer: ______________________

**5** Solve for x and find MN:

$MN = x + 5, \quad b_1 = 2x + 2, \quad b_2 = x - 2$

Answer: ______________________

**6** Solve for x and find MN:

$MN = 3x - 1, \quad b_1 = 4x + 2, \quad b_2 = x + 4$

Answer: ______________________

**7** Solve for x. Find b1 and b2:

$MN = 12, \quad b_1 = 3x - 4, \quad b_2 = x + 4$

Answer: ______________________

**8** Solve for x. Find MN, b1, and b2:

$MN = 3x + 2, \quad b_1 = 5x - 4, \quad b_2 = 2x + 6$

Answer: ______________________

# Answer Key & Solutions

*Geometry: Trapezoid Midsegment Theorem • Numberbender*

**TEACHER NOTES** *Midsegment Theorem: MN = (b1+b2)/2, or equivalently 2·MN = b1+b2. MN is parallel to both bases. To find a missing base: b2 = 2·MN – b1.*

**1** Find MN:

$b_1 = 7, \quad b_2 = 15$

= $MN = 11$

*MN = (b1+b2)/2 = (7+15)/2 = 11.*

**2** Find MN:

$b_1 = 5, \quad b_2 = 23$

= $MN = 14$

*MN = (5+23)/2 = 14.*

**3** Find b2:

$MN = 13, \quad b_1 = 9$

= $b_2 = 17$

*13=(9+b2)/2 → 26=9+b2 → b2=17.*

**4** Find b1:

$MN = 16, \quad b_2 = 10$

= $b_1 = 22$

*16=(b1+10)/2 → 32=b1+10 → b1=22.*

**5** Solve for x and find MN:

$MN = x+5, \quad b_1 = 2x+2, \quad b_2 = x-2$

= $x = 10, \quad MN = 15$

*x+5=(2x+2+x–2)/2=3x/2 → 2x+10=3x → x=10. MN=15.*

**6** Solve for x and find MN:

$MN = 3x-1, \quad b_1 = 4x+2, \quad b_2 = x+4$

= $x = 8, \quad MN = 23$

*3x–1=(4x+2+x+4)/2=(5x+6)/2 → 6x–2=5x+6 → x=8. MN=23.*

**7** Solve for x. Find b1 and b2:

$MN = 12, \quad b_1 = 3x-4, \quad b_2 = x+4$

= $x = 6, \quad b_1 = 14, \quad b_2 = 10$

*12=(3x–4+x+4)/2=4x/2=2x → x=6. b1=14, b2=10.*
*Check:(14+10)/2=12 ✓*

**8** Solve for x. Find MN, b1, and b2:

$MN = 3x+2, \quad b_1 = 5x-4, \quad b_2 = 2x+6$

= $x = 2, \quad MN = 8, \quad b_1 = 6, \quad b_2 = 10$

*3x+2=(5x–4+2x+6)/2=(7x+2)/2 → 6x+4=7x+2 → x=2.*
*Check:(6+10)/2=8 ✓*